1. If $x + \frac{1}{x} = 2$, then x equals

A) 1

B) -1

C) 0

D) 2

Answer: A

---

2. The value of $\sin^2\theta + \cos^2\theta$is

A) 0

B) 1

C) 2

D) -1

Answer: B

---

3. If $\log a + \log b = 1$, then ab =

A) 1

B) 10

C) 0

D) 100

Answer: B

---

4. The number of real roots of $x^2 + 1 = 0$is

A) 0

B) 1

C) 2

D) Infinite

Answer: A

---

5. If determinant of a matrix is zero, then matrix is

A) Identity

B) Singular

C) Diagonal

D) Unit

Answer: B

---

6. The derivative of $x^2$is

A) x

B) 2x

C) x^2

D) 1

Answer: B

---

7. The integral of 1 dx is

A) x

B) x^2

C) 1

D) 0

Answer: A

---

8. The limit of $\frac{\sin x}{x}$as x→0 is

A) 0

B) 1

C) ∞

D) -1

Answer: B

---

9. The slope of line $y = mx + c$is

A) c

B) m

C) x

D) y

Answer: B

10. Distance between (0,0) and (3,4) is

A) 5

B) 6

C) 7

D) 4

Answer: A

---

11. If $x^2 - 5x + 6 = 0$, roots are

A) 2,3

B) 1,6

C) 3,4

D) 2,4

Answer: A

---

12. The sum of first n natural numbers is

A) n^2

B) n(n+1)/2

C) n(n-1)/2

D) n^3

Answer: B

---

13. The value of $\tan 45°$is

A) 0

B) 1

C) √3

D) ∞

Answer: B

---

14. If vectors are perpendicular, dot product is

A) 1

B) 0

C) -1

D) ∞

Answer: B

---

15. Probability of certain event is

A) 0

B) 1

C) 1/2

D) -1

Answer: B

---

16. If $x^2 = 9$, x equals

A) 3

B) -3

C) ±3

D) 0

Answer: C

---

17. The domain of $\sqrt{x-2}$is

A) x≥2

B) x≤2

C) x>2

D) x<2

Answer: A

---

18. The range of sin x is

A) [-1,1]

B) [0,1]

C) (-∞,∞)

D) [0,∞)

Answer: A

---

19. The number of permutations of 3 objects is

A) 3

B) 6

C) 9

D) 12

Answer: B

---

20. The number of combinations of 3 objects taken 2 is

A) 3

B) 6

C) 2

D) 1

Answer: A

---

21. The equation of circle with center (0,0) is

A) $x^2+y^2=r^2$

B) $x+y=r$

C) $x^2-y^2=r^2$

D) $xy=r$

Answer: A

---

22. The eccentricity of parabola is

A) 0

B) 1

C) >1

D) <1

Answer: B

---

23. The value of $i^2$is

A) 1

B) -1

C) i

D) 0

Answer: B

---

24. The modulus of complex number is always

A) Negative

B) Zero

C) Positive

D) Imaginary

Answer: C

---

25. If $f(x)=x^3$, then $f'(x)=$

A) x^2

B) $3x^2$

C) x^3

D) $3x$

Answer: B

---

26. If determinant is non-zero, system is

A) No solution

B) Unique solution

C) Infinite

D) Undefined

Answer: B

---

27. The mean of data is

A) Sum/n

B) Product

C) Difference

D) Ratio

Answer: A

---

28. Standard deviation measures

A) Central value

B) Spread

C) Mean

D) Median

Answer: B

---

29. The value of cos 0° is

A) 0

B) 1

C) -1

D) ∞

Answer: B

---

30. The derivative of sin x is

A) cos x

B) -cos x

C) sin x

D) -sin x

Answer: A

---

31. The integral of cos x dx is

A) sin x

B) -sin x

C) cos x

D) -cos x

Answer: A

---

32. The value of $\sec^2 x - \tan^2 x$ is

A) 1

B) 0

C) -1

D) 2

Answer: A

---

33. If $A \subset B$, then $n(A)$

A) $\leq n(B)$

B) $\geq n(B)$

C) $= n(B)$

D) None

Answer: A

---

34. If vectors are parallel, cross product is

A) 0

B) 1

C) ∞

D) -1

Answer: A

---

35. The derivative of ln x is

A) x

B) 1/x

C) ln x

D) e^x

Answer: B

---

36. The value of e^0 is

A) 0

B) 1

C) e

D) ∞

Answer: B

---

37. The equation $x^2+y^2=0$ represents

A) Point

B) Circle

C) Line

D) Pair

Answer: A

---

38. The slope of vertical line is

A) 0

B) ∞

C) 1

D) -1

Answer: B

---

39. The determinant of identity matrix is

A) 0

B) 1

C) n

D) -1

Answer: B

---

40. If A+B=B+A, matrices are

A) Non-commutative

B) Commutative

C) Symmetric

D) Singular

Answer: B

41. The number of subsets of a set with n elements is

A) n^2

B) 2^n

C) n!

D) n

Answer: B

---

42. If probability of event is 0, it is

A) Sure

B) Impossible

C) Random

D) Equal

Answer: B

---

43. The value of tan 90° is

A) 0

B) 1

C) ∞

D) -1

Answer: C

---

44. The derivative of constant is

A) 0

B) 1

C) Constant

D) ∞

Answer: A

---

45. If $x \to \infty$, $1/x \to$

A) 1

B) 0

C) ∞

D) -1

Answer: B

---

46. The sum of GP is

A) a/(1-r)

B) a(1-r)

C) ar

D) a+r

Answer: A

---

47. The condition for parallel lines is

A) Same slope

B) Different slope

C) Opposite slope

D) Zero slope

Answer: A

---

48. The condition for perpendicular lines is

A) $m_1m_2 = -1$

B) $m_1=m_2$

C) $m_1+m_2=1$

D) None

Answer: A

---

49. The value of sin 90° is

A) 0

B) 1

C) -1

D) ∞

Answer: B

---

50. The inverse of matrix exists if determinant is

A) 0

B) ≠0

C) 1

D) -1

Answer: B

---

51. The distance formula is

A) $\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2]}$

B) x+y

C) x-y

D) xy

Answer: A

---

52. The derivative of e^x is

A) e^x

B) x

C) 1

D) ln x

Answer: A

---

53. The integral of x dx is

A) $x^2/2$

B) x

C) 1

D) x^3

Answer: A

---

54. The value of $\log_{10}1$ is

A) 0

B) 1

C) 10

D) -1

Answer: A

---

55. If two events are independent, $P(A \cap B)=$

A) $P(A)+P(B)$

B) $P(A)P(B)$

C) $P(A)-P(B)$

D) None

Answer: B

---

56. The mean of symmetric distribution is

A) Equal to median

B) Zero

C) Infinite

D) None

Answer: A

---

57. The value of cos 90° is

A) 0

B) 1

C) -1

D) ∞

Answer: A

---

58. The equation of straight line is

A) $ax+by+c=0$

B) $x^2+y^2=0$

C) $xy=0$

D) $x^3=0$

Answer: A

---

59. The derivative of tan x is

A) $\sec^2 x$

B) cos x

C) sin x

D) tan x

Answer: A

---

60. The integral of $\sec^2 x\, dx$ is

A) tan x

B) cot x

C) sec x

D) cos x

Answer: A